

MBAI 5310G-001: AI Programming

Assignment 4: Decision Tree Model

Evergreen Bank: Credit Card Upgrade

Field	Detail
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1. Business Overview

Evergreen Bank is a mid-sized Canadian financial institution that serves over 200,000 personal banking customers. The bank offers a tiered credit card product line, and management has identified a significant revenue opportunity in upgrading existing cardholders to a premium tier, specifically the Evergreen Platinum Card, which carries higher annual fees, better rewards, and increased credit limits.

Rather than send the upgrade offer to every customer, the bank wants to identify in advance which customers are most likely to accept. This avoids marketing waste and reduces the risk of annoying customers who are likely to decline. The assignment applies a Decision Tree classification model to predict upgrade acceptance using customer profile, banking relationship, card usage, digital engagement, and credit risk features.

Attribute	Detail
Business	Evergreen Bank
Objective	Predict which customers will accept a credit card upgrade offer
Target Variable	Accepted_Card_Upgrade (Yes / No)
Dataset Size	340 synthetic customer records
Number of Features	20 input features
Priority Metric	Precision, minimize false positives (wasted offers)
Model Used	Decision Tree Classifier (sklearn, random_state=42)

2. Data Preparation

2.1 Dataset Overview

The dataset contains 340 records representing individual Evergreen Bank customers. Each record has 20 features spanning five categories:

- Customer Profile: Age_Group, Gender, Region, Income_Band, Employment_Status
- Banking Relationship: Tenure_Years, Num_Products, Primary_Account_Type
- Card Usage: Credit_Utilization_Pct, Travel_Spend_Share_Pct, Dining_Spend_Share_Pct, Monthly_Spend_Avg
- Digital Engagement: App_Logins_Per_Month, Online_Banking_User, Paperless_Billing
- Credit Risk: Credit_Score_Band, Missed_Payments_Last_12M, Debt_to_Income_Ratio

2.2 Preprocessing Steps

The following preprocessing steps were applied before model training:

- Dropped Customer_ID, an identifier with no predictive value
- Checked for and confirmed no missing values in the dataset
- Confirmed no duplicate records

- Applied one-hot encoding (pd.get_dummies) to all categorical features
- Applied Label Encoding (LabelEncoder) to the target variable: Yes = 1, No = 0
- Split data 80/20 using train_test_split with stratify=y and random_state=42

3. Decision Tree Model

3.1 How Decision Trees Make Predictions

A Decision Tree is a supervised machine learning algorithm that learns a series of yes/no questions about the input features. Starting from the root node, the algorithm selects the feature that best separates the classes at each step (using Gini impurity or information gain). The process continues until the tree reaches a leaf node, which outputs a final class prediction.

The primary advantage of Decision Trees is interpretability, you can trace exactly why a particular prediction was made by following the path from root to leaf. This is especially valuable in a financial services context where decisions affecting customers must be explainable.

3.2 Decision Tree Visualization

The tree below shows the first three levels of the trained decision tree. Each node shows the splitting feature, Gini impurity, sample count, and class distribution. Nodes coloured orange indicate the model predicts No (customer declines the upgrade); nodes coloured blue indicate Yes (customer accepts). Darker shading indicates a purer node, a stronger majority of one class.

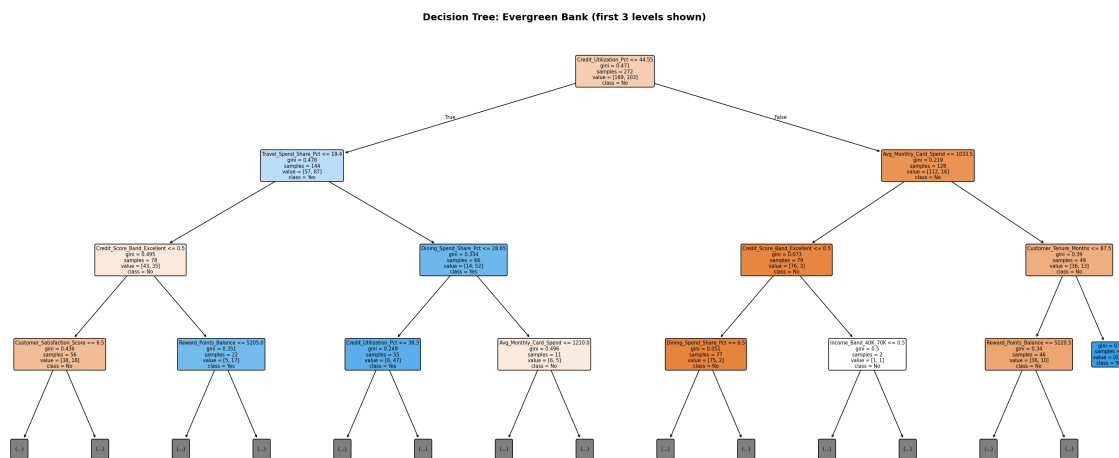


Figure 1: Decision Tree, Evergreen Bank (first 3 levels shown)

4. Model Evaluation

4.1 Confusion Matrix

The confusion matrix below shows the breakdown of correct and incorrect predictions on the test set (68 records). The rows represent the actual class, what the customer actually did. The columns represent the predicted class, what the model said they would do.

Confusion Matrix: Evergreen Bank (unconstrained tree)

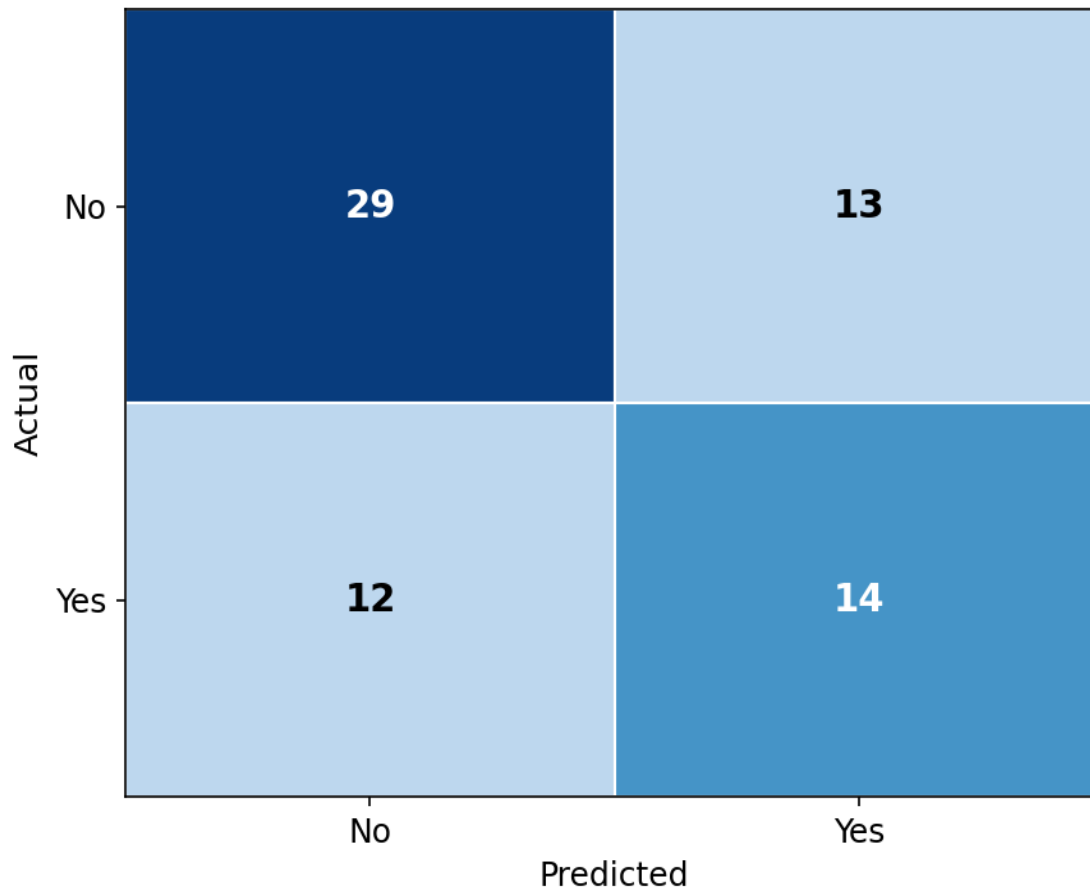


Figure 2: Confusion Matrix, Evergreen Bank (unconstrained tree)

Reading the matrix:

- Top-left (29) - True Negatives: The model correctly predicted 29 customers would decline the upgrade, and they did. No offer was wasted.
- Top-right (13) - False Positives: The model predicted 13 customers would accept, but they actually declined. These are wasted upgrade offers, marketing cost with no return.
- Bottom-left (12) - False Negatives: The model predicted 12 customers would decline, but they would have accepted. These are missed revenue opportunities, customers who wanted the upgrade but were never offered it.
- Bottom-right (14) - True Positives: The model correctly identified 14 customers who accepted the upgrade. These are the successful targeted conversions.

For Evergreen Bank, false positives are the more costly error. Sending an upgrade offer to a customer who will decline wastes marketing budget and risks annoying a customer. This is why Precision is the priority metric; it directly measures how often a predicted 'Yes' is actually correct.

4.2 Classification Metrics

Metric	Result
Training Accuracy	100.0%
Testing Accuracy	63.2%
Precision (Yes: accepted upgrade)	51.9%
Recall (Yes: accepted upgrade)	53.8%
F1-Score (Yes: accepted upgrade)	52.8%
False Positives (offered but would decline)	13
False Negatives (not offered but would accept)	12

4.3 Overfitting Analysis

The unconstrained tree achieves 100% accuracy on training data but only 63.2% on the test set, a gap of approximately 36.8 percentage points. This is a clear sign of overfitting: the model memorized the training examples rather than learning patterns that generalize to new customers.

Overfitting is particularly risky in a banking context. A model that appears highly accurate internally but performs poorly on real customers would lead to poor targeting decisions and wasted marketing spend.

5. Controlling Complexity with max_depth

5.1 Applying max_depth=4

To address overfitting, the model was retrained with max_depth=4. This restricts the tree to a maximum of 4 levels of splits, preventing it from learning overly specific rules that don't generalize. The result is a simpler, more robust model.

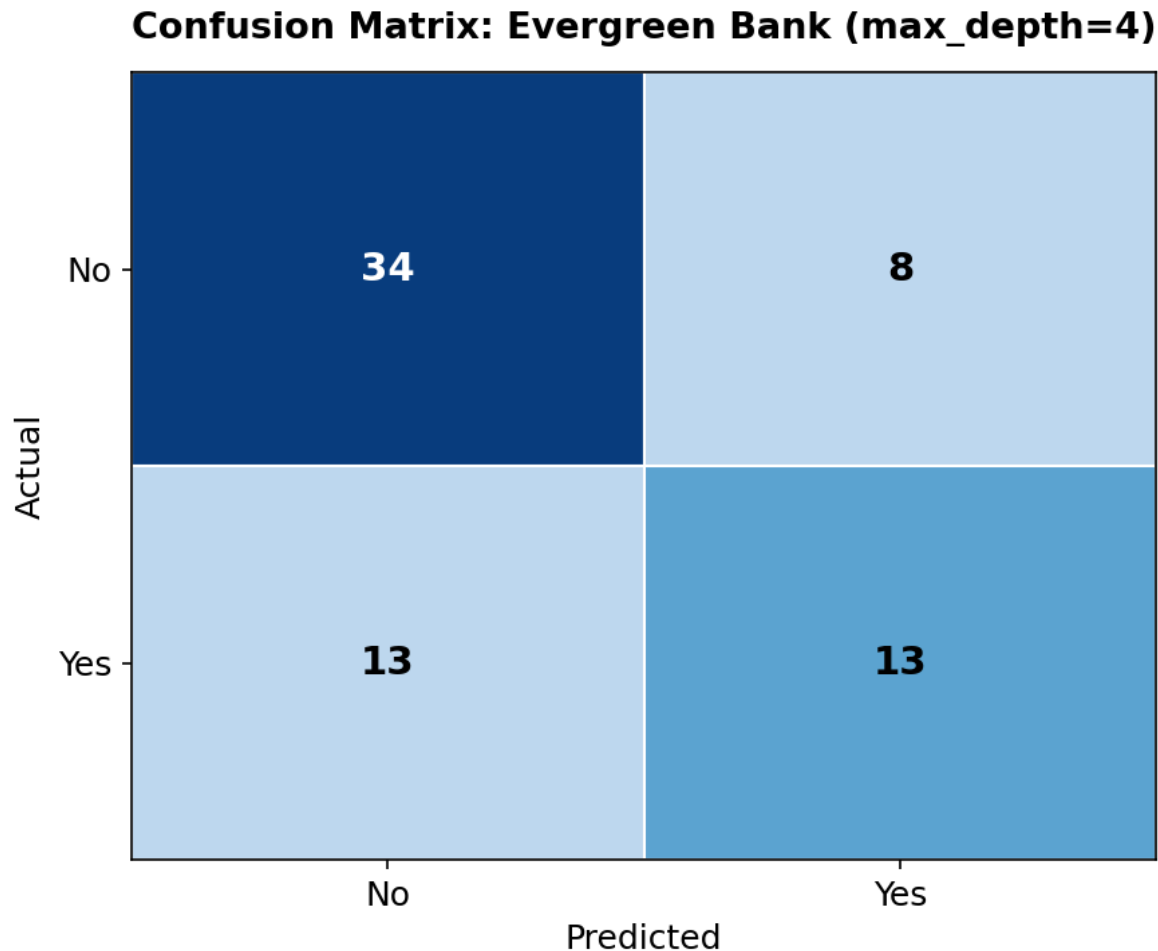


Figure 3: Confusion Matrix, Evergreen Bank (max_depth=4)

Reading the max_depth=4 matrix:

- True Negatives: Customers correctly predicted to decline, the model avoided sending them a wasted offer.
- False Positives: Customers predicted to accept who actually declined, still a source of marketing waste, but reduced compared to the unconstrained model.
- False Negatives: Customers predicted to decline who would have accepted, missed upgrade opportunities.
- True Positives: Customers correctly identified as likely to accept, the core value of the model.

With max_depth=4, testing accuracy improved from 63.2% to 69.1%. The model makes fewer extreme errors by avoiding the overly specific rules the unconstrained tree learned, rules that worked on training data but failed on new customers.

Comparison	Unconstrained
Training Accuracy	100.0%
Testing Accuracy	63.2%
Train/Test Gap	~36.8 pts

Comparison	Unconstrained	max_depth=4
Training Accuracy	100.0%	—
Testing Accuracy	63.2%	69.1%
Train/Test Gap	~36.8 pts	~30.9 pts

6. Feature Importance

After applying max_depth=4, the model identified the following as the most influential features in predicting whether a customer will accept the upgrade offer. Feature importance scores reflect how much each feature reduces impurity across all splits in the tree.

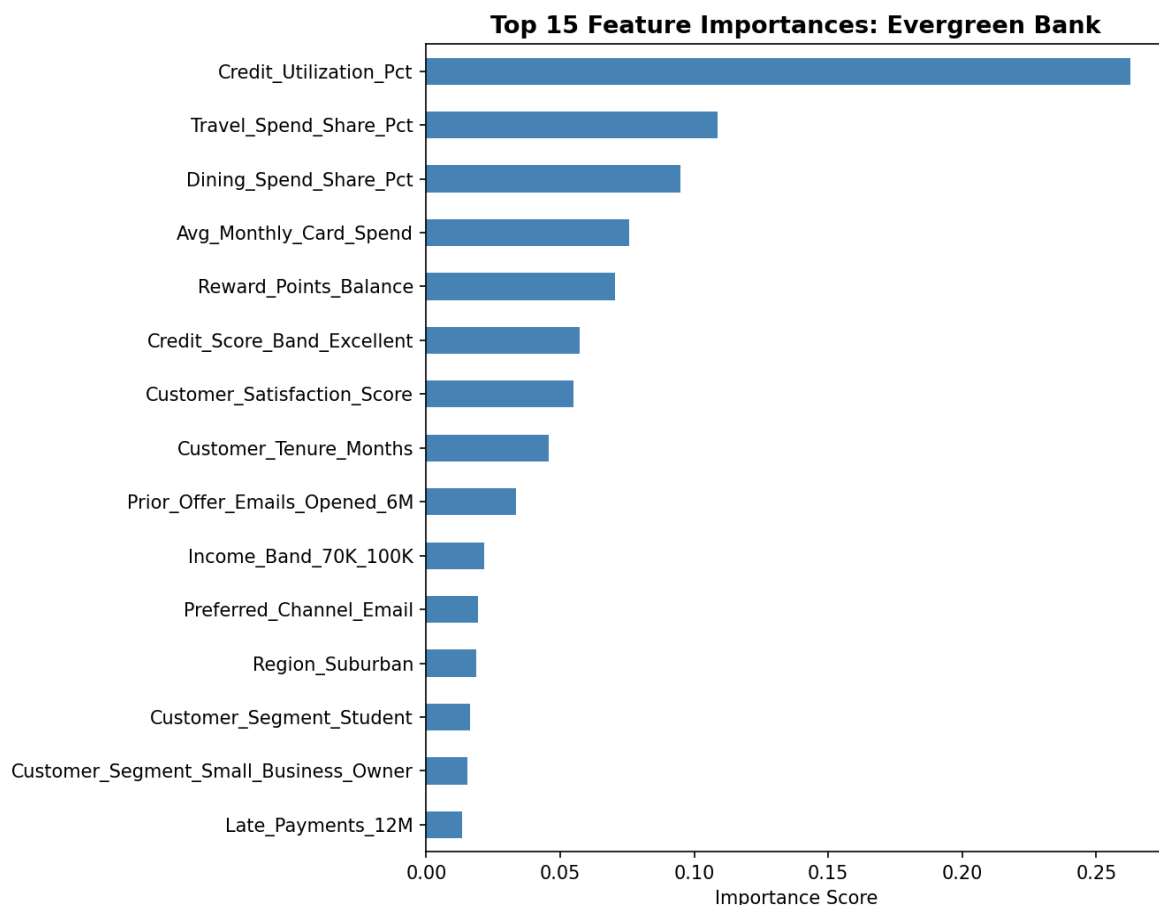


Figure 4: Top 15 Feature Importances, Evergreen Bank

6.1 Business Interpretation of Top Features

Feature	Business Interpretation
Credit_Utilization_Pct	Customers using 30-70% of their credit limit show interest in credit products without being overextended, a strong upgrade signal.
Travel_Spend_Share_Pct	High travel spenders are drawn to premium cards with travel rewards, lounge access, and travel insurance.
Dining_Spend_Share_Pct	Lifestyle spenders (dining, entertainment) align with the premium card's rewards structure.
Avg_Monthly_Card_Spend	Higher average monthly card spend indicates active card usage and a lifestyle that would benefit from premium rewards and higher limits.
Reward_Points_Balance	Customers with accumulated reward points are already engaged with the card's benefits program, a strong signal of readiness to upgrade to a more rewarding tier.

7. Business Actions and Recommendations

Model Insight	Recommended Business Action
Credit utilization is the top predictor	Target customers in the 30-70% utilization band for upgrade outreach, they are credit-active but not over-leveraged.
Travel and dining spend drive acceptance	Tailor upgrade offer messaging around travel rewards, airport lounge access, and dining cashback to match what these customers already value.
High monthly spenders are better candidates	Prioritize outreach to customers with above-average monthly spend; offer them an upgrade trial period or fee waiver.
Model precision is 51.9%	Nearly half of predicted acceptors will decline. Pair model output with customer service review before sending offers to borderline cases.
max_depth=4 improved test accuracy to 69.1%	Retrain and tune the model quarterly as new customer spending and behavior data comes in to maintain and improve accuracy.

8. Limitations

The following limitations should be considered before deploying this model in a real banking environment:

- Synthetic dataset: The 340-record dataset was synthetically generated for educational purposes. Real-world performance will differ significantly with actual customer data.
- Small sample size: 340 records is insufficient for a robust production model. A minimum of several thousand records would be needed for reliable generalization.
- Demographic features risk bias: Features such as Income_Band, Age_Group, Gender, and Region may introduce demographic bias. For example, the model could systematically under-offer upgrades to lower-income or younger customers, even when they would genuinely benefit.
- Static model: Customer behavior changes over time. A model trained on historical data will degrade in accuracy without regular retraining.
- Binary target only: The model predicts only accept/decline. It does not capture the probability of acceptance, which would be more useful for ranking and prioritizing outreach.

9. Responsible AI Considerations

Financial services AI applications carry a high responsibility to treat customers fairly. The following considerations apply to this model:

- Fairness review required: Before deployment, the model must be audited for disparate impact across demographic groups, particularly Income_Band, Age_Group, Gender, and Region. Systematically excluding any protected group from upgrade offers would raise regulatory and ethical concerns.
- Human oversight: All upgrade offer decisions should be reviewed by a bank representative before being sent, particularly for borderline predictions. The model should inform, not automate, the final decision.
- Explainability: Evergreen Bank must be able to explain to regulators and customers why a particular offer was or was not made. The Decision Tree's interpretable structure supports this, but the specific decision path should be logged for each customer prediction.
- Data privacy: Customer financial and behavioral data used in model training must be handled in accordance with PIPEDA (Canada's federal privacy law) and any applicable provincial regulations.
- Consent: Customers should be informed that their data is being used for predictive modeling purposes, in line with the bank's privacy policy.

10. Final Conclusion

This assignment applied a Decision Tree classifier to a realistic banking problem, predicting which Evergreen Bank customers are likely to accept a premium credit card upgrade. The model was trained on 340 synthetic customer records using 20 input features, with an 80/20 stratified train/test split.

The unconstrained tree achieved 100% training accuracy but only 63.2% test accuracy, confirming severe overfitting. Applying `max_depth=4` reduced this gap and improved test accuracy to 69.1%. The most influential predictors were `Credit_Utilization_Pct`, `Travel_Spend_Share_Pct`, and `Dining_Spend_Share_Pct`, all of which reflect customer lifestyle and engagement patterns that intuitively align with premium card interest.

Precision was selected as the priority metric because false positives, sending upgrade offers to customers who will decline, waste marketing resources and create a poor customer experience. At 51.9% precision, the model shows promise but requires further tuning, more data, and a fairness review before it could be deployed in a real banking environment.

Decision Trees are a strong starting point for this type of problem due to their interpretability. Future work should explore ensemble methods such as Random Forest or Gradient Boosting to improve predictive accuracy while maintaining explainability.